

Jiangming Yao

Curriculum vitae

Professor
School of Physics and Astronomy, Sun Yat-sen University
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Research Interest

- Quantum many-body approaches: ab initio calculations, in-medium similarity renormalization group, chiral effective field theory, nuclear density functional theory
- Modeling nuclear matrix elements for new physics: nuclear Schiff moments and CP violation, neutrinoless double beta decay and neutrino physics
- Application to nuclear structure, hypernuclei, nuclear matter and neutron stars

Professional experience

- 2021–Now **Professor**, School of Physics and Astronomy, Sun Yat-sen University, China
2026–2026 **Visiting Professor**, YITP, Kyoto University, Japan
2018–2021 **Postdoctoral Research Associate**, FRIB/NSCL, Michigan State University, USA
2015–2017 **Postdoctoral Research Associate**, UNC at Chapel Hill, USA
2013–2015 **Assistant Professor**, Tohoku University, Japan
2011–2012 **Postdoc**, Université Libre de Bruxelles, Belgium
2009–2017 **Professor**, Southwest University, China

Education

- 2004–2009 **PhD in Nuclear and Particle Physics**, Peking University, China
Thesis: Covariant density functional theory for nuclear spectroscopy
Supervisor: Jie Meng
2006–2008 **Exchange PhD program**, Technical University of Munich, Germany
Supervisor: Peter Ring
2000–2004 **BSc**, Nankai University, China

Teaching and tutoring experience

- 2022–now **Advanced Quantum Mechanics**, graduate course, Sun Yat-sen University, China
2022–now **Atomic Physics**, undergraduate course, Sun Yat-sen University, China
2021–now **Nuclear Theory**, graduate course, Sun Yat-sen University, China
2021–now **Introduction to Nuclear and Particle Physics**, undergraduate course, Sun Yat-sen University, China
2005 **Teaching Assistant**, Peking University, China

2009–2011 **Modern physics for undergraduate students**, *Southwest University, China*
Nuclear Theory for graduate students, *Southwest University, China*

Master Students Mentored

- Jian Xiang **2009-2012**, *Covariant density functional theory for nuclear shape evolution and octupole vibration*, Master Thesis of Southwest University, Chongqing
- Yun Fu **2010-2013**, *Covariant density functional theory for Kr isotopes and relativistic projected shell model*, Master Thesis of Southwest University, Chongqing
- Xianye Wu **2011-2014**, *Systematic study of low-lying states in light nuclei with multi-reference covariant density functional theory*, Master Thesis of Southwest University, Chongqing
- Weixia Xue **2012-2015**, *Covariant density functional theory for Lambda impurity effect in nuclear spectroscopy*, Master Thesis of Southwest University, Chongqing
- Enfu Zhou **2013-2016**, *Collective excitations of octupole deformed nuclei with multi-reference covariant density functional theory*, Master Thesis of Southwest University, Chongqing
- Wei Lin **2021-2024**, *The study of low-lying states of odd-mass nuclei with nuclear chiral interaction*, Master Thesis of Sun Yat-sen University, Zhuhai
- Xin Zhang **2021-2024**, *Application of Machine Learning and Eigenvector Continuation to Generator Coordinate Method for Nuclear Physics*, Master Thesis of Sun Yat-sen University, Zhuhai

PhD students Mentoring

- Enfu Zhou **2022-2025**, *Multi-reference covariant density functional theory for nuclear Schiff moments*
- Ruonan Chen **2023-2026**, *Symmetry-restored quasiparticle random-phase approximation for nuclear beta decay*
- Chenrong Ding **2023-2027**, *Ab initio studies of nuclear matrix elements of neutrinoless double-beta decay in heavy candidate nuclei*

Postdocs Mentored

- Chen-can Wang **2022-2024**, *Relativistic ab initio studies of nuclear matter and finite nuclei*

Prizes and awards

- 2023 **J. M. Yao (PI)**, *National Natural Science Foundation of China*, Ab initio description of medium-mass deformed nuclei with in-medium generator coordinate method, Grant No. 12375119
\$70,000 for period 01/01/2024–12/30/2027

- 2022 **J. M. Yao (PI)**, *Guangdong Basic and Applied Basic Research Foundation*, First-Principles Calculation of Nuclear Matrix Elements for Neutrinoless Double Beta Decay in Atomic Nuclei, Grant No. 2023A1515010936
\$15,000 for period 01/01/2023–12/30/2025
- 2021 **J. M. Yao (Co-PI)**, *National Natural Science Foundation of China*, Theoretical Study on Neutrinoless Double-Beta Decay, Grant No. 12141501
\$300,000 for period 01/01/2022–12/30/2025
- 2015 **J. M. Yao (PI)**, *National Natural Science Foundation of China*, Multi-reference covariant energy density functional theory for odd-odd nuclei, Grant No. 11575148
\$80,000 for period 01/01/2016–12/30/2019
- 2014 **J. M. Yao (Co-PI)**, *Grant-in-Aid for Scientific Research (C)*, Low-lying collective excitations of hypernuclei with microscopic particle-rotor model, Grant No. 26400263
4,550,000 JPY for period 04/01/2014 –03/31/2017
- 2013 **J. M. Yao (PI)**, *National Natural Science Foundation of China*, Covariant energy density functional theory for the hyperon impurity effect in atomic nuclei, Grant No. 111105111
\$30,000 for period 01/01/2012 –12/30/2014
- 2009 **J. M. Yao (PI)**, *National Natural Science Foundation of China*, Covariant energy density functional theory for the low-lying states of exotic nuclei, Grant No. 10947013
\$20,000 for period 01/01/2010–12/30/2012
- 2006-2008 **European Community project Asia-Europe Link in Nuclear Physics and Astrophysics**, CN/ASIA-LINK/008 (094-791)
- 2008 **Wu-Si scholarship for top students**, *Peking University*
- 2001 **First prize in the mathematics competition for university students**, *Tianjin municipality*

Professional service

- Reviewer for **National Natural Science Foundation of China**
Academic Degree and Education Evaluation of Chinese Ministry of Education
- Reviewer for **Physical Review (Letters, C)**, **Nature communication**, **Science Bulletin**,
Journals **Progress of Theoretical and Experimental Physics**, **Chinese Physics C**, **European Physical Journal A**, **International Journal of Modern Physics E**, **Communication in Theoretical Physics**, **Frontiers of Physics**, **Central European Journal of Physics**, **Science in China: Physics, Mechanics and Astronomy**, **Science Bulletin**, etc
- Reviewer **Frontier in Physics (Nuclear Physics)**, **Symmetry**
editor for

Organizers of conferences and workshops

- Co-organizer **The long-term workshop on "Intersection of nuclear structure and high-energy nuclear collisions 2026"**, 13–24 Apr 2026 **Yukawa Institute for Theoretical Physics**, *link*, 2026

- Chair **The workshop on "Nuclear ab initio calculations and frontiers of beta decays", 2-4, Jan, 2025, Zhuhai, China, [link](#), 2025**
- Chair **The workshop on "Generator coordinate method in nuclear physics", Zhuhai, March 14-16, 2024**
- Co-organizer **The 2nd Symposium on Nuclear Physics in Guangdong-Hong Kong-Macao Area, Guangzhou, June 17-20, [link](#), 2023**
- Co-organizer **The 2nd workshop on "Neutrinoless double-beta decay and related topics", Zhuhai, May 18-23, [link](#), 2023**
- Chair **The 1st Symposium on Nuclear Physics in Guangdong-Hong Kong-Macao Area, Zhuhai, July 2-6, [link](#), 2022**
- Chair **The 1st workshop on "Neutrinoless double-beta decay and related topics", Zhuhai, May 18-23, [link](#), 2021**

References

Kouichi Hagino, hagino.kouichi.5m@kyoto-u.ac.jp

Heiko Hergert, hergert@frib.msu.edu

Jie Meng, mengj@pku.edu.cn

Jonathan Engel, engelj@physics.unc.edu

Peter Ring, peter.ring@tum.de

Publications (see publication list for details)

- Summary **100+ peer reviewed research papers**
20+ conference proceedings
4 book chapters
H-index is 33 and total number of citations is 3,000+ , *Web of Science*
- ResearcherID **<https://publons.com/researcher/1544854/jiangming-yao/>**
- Inspirehep **http://inspirehep.net/search?ln=zh_CN&ln=zh_CN&p=find+au+j.+m.+yao&of=hcs&action_search=??&sf=&so=d&rm=&rg=25&sc=0**
- ORCID **<http://orcid.org/0000-0001-9505-1852>**

Invited talks in conferences/workshops

- 2025/12/13 **Modeling Nuclear Weak Processes Based on Operators from Chiral Effective Field Theory**, *The First National Workshop on Nuclear Astrophysics, Shenzhen, China*
- 2025/11/30 **Nuclear Matrix Elements for $0\nu\beta\beta$ Decay**, *Workshop on Neutrinos and Related New Physics Research (NuPhyR), Zhuhai, China, [link](#)*
- 2025/10/30 **Advances in Modeling Nuclear Matrix Elements for New Physics Searches**, *The 15th Frontiers Forum on Relativistic Density Functional Theory, Luoyang, China, [link](#)*

- 2025/10/25 **Recent Advances in Ab Initio Modeling Nuclear Matrix Elements for Weak Processes: Progress and Perspectives**, *The Second Workshop on Neutrino Scattering: Theory, Experiment and Phenomenology (vSTEP 2025)*, Beijing, China, link
- 2025/10/18 **Advances in Modeling Nuclear Matrix Elements for New Physics Research**, *The 10th Workshop on Chiral Effective Field Theory*, Nanjing, China, link
- 2025/06/27 **Multi-Reference Covariant Density Functional Theory for Nuclear Schiff Moments**, *Workshop on New Generation Nuclear Density Functionals 2025*, Beijing, China, link
- 2025/05/18 **Modeling Nuclear Neutrinoless Double-Beta Decay: Progress and Challenges in Nuclear Physics**, *Intersection of Nuclear Structure and High-Energy Nuclear Collisions 2025*, Shanghai, China, link
- 2025/05/12 **Beyond Mean-Field Correlations on Nuclear Schiff Moments and P,T-Odd Nuclear Forces**, *Electric Dipole Moments: Experimental and Theoretical Horizons*, online, link
- 2025/03/25 **Transition from Spin Symmetry to Pseudospin Symmetry in Nuclear Shell Structure: A Perspective from Realistic Nuclear Forces**, *Seminar, Institute of Theoretical Physics, Chinese Academy of Sciences*, Beijing, China, link
- 2025/03/15 **Multireference CDFT for Nuclear Schiff Moments—A Test of Fundamental Symmetry**, *The 8th International Workshop on DRHBc Mass Table*, The University of Hong Kong, Hong Kong, China, link
- 2024/12/20 **Advances in modeling nuclear matrix elements for $0\nu\beta\beta$ decay**, *Workshop on Double Beta Decay and Related Underground Experiments*, Dec.20-21, Tsung-Dao Lee Institute, Shanghai, China , (Link)
- 2024/11/16 **Modeling nuclear matrix elements for advancing new physics research in the era of high precision**, *The 2024 Asian Nuclear Physics Association Symposium*, Nov. 16, 2024, Huizhou, Guangdong, China , (Link)
- 2024/10/19 **Modeling $0\nu\beta\beta$ decay based on nuclear forces and transition operators from chiral effective field theory**, *The 9th workshop on chiral effective field theory*, Changsha, Hunan, China, Oct. 19 , (Link)
- 2024/10/14 **Advances in the modeling of nuclear matrix elements of neutrinoless double-beta decay based on non-relativistic chiral nuclear forces**, *The 14th symposium on relativistic density functional*, Nanjing, Jiangsu, China, Oct.14
- 2024/5/25 **Multi-Reference In-Medium Similarity Renormalization Group Method and its Applications: Nuclear Shell Structure and Low-Lying States**, *Workshop on 'Nuclear Shell Model and Frontiers of Nuclear Physics'*, China Institute of Atomic Energy, Institute of Nuclear Physics, Beijing, May 24-27, (Link)
- 2024/5/22 **Low-lying spectroscopy of even-even nuclei via projected generator coordinate method calculations with a multi-reference in-medium similarity renormalization group pre-processing of the Hamiltonian**, *Workshop of the Espace de Structure et de réactions Nucléaires Théorique: Nuclear ab initio spectroscopy*, CEA, Orme des Merisiers Campus, 91191 Gif-sur-Yvette, France, May 21-24, (Link)

- 2024/5/18 **Nuclear Matrix Elements for $0\nu\beta\beta$ Decay: Progress and Prospects**, *The First Workshop on Neutrino Scattering: Theory, Experiment, Phenomenology (vSTEP 2024)*, Hangzhou Institute for Advanced Study, University of Chinese Academy of Sciences, Hangzhou, May 17-20, (Link)
- 2024/4/27 **Advances in ab initio studies of deformed nuclei and neutrinoless double-beta decay with renormalization group methods**, *Long-term workshop on "Exploring nuclear physics across energy scales"*, Beijing, April 15-28, (Link)
- 2023/12/16 **Advances in Modeling neutrinoless double-beta decay with operators from chiral effective field theory**, *Annual meeting of NuDEx-CUPID-China*, Huizhou, Dec. 15-17, 2023
- 2023/12/21 **Modeling neutrinoless double-beta decay with operators from chiral effective field theory**, *The international workshop on the theoretical and experimental approaches for nuclear matrix elements of double-beta decay*, Research Center for Nuclear Study, Osaka University, Osaka, Dec. 21-22, (Link)
- 2023/11/18 **Multireference covariant density functional theory for low-lying states of odd-mass nuclei**, *Special workshop on 'Relativistic First-Principles Calculations of Atomic Nuclei'*, China Institute of Atomic Energy, Institute of Nuclear Physics, Beijing, Nov. 17-19, (Link)
- 2023/10/28 **Modeling neutrinoless double-beta decay with operators from chiral effective field theory**, *The 8th workshop on chiral effective field theory*, Kaifeng, Henan, China, Oct. 27-31
- 2023/5/18 **Ab initio calculations of nuclear matrix elements for neutrinoless double beta decay**, *The NuDEx Collaboration meeting*, Institute of Modern Physics, Chinese Academy of Science, Lanzhou, Mar. 17-19, 2023
- 2022/05/26 **Advances in modeling nuclear matrix elements of neutrinoless double beta decay**, *The Forum on High Energy Nuclear Physics in China*, May 26, 2022
- 2021/12/18 **Advances in modeling medium-mass hypernuclei**, *the 4th International Workshop on Strangeness Nuclear Physics (SNP)*, Dec 18-19, 2021
- 2018/06/26 **Beyond relativistic mean-field approach to deformed hypernuclei**, *The 13th International Conference on Hypernuclear and Strange Particle Physics*, June 24-29, 2018, Portsmouth Virginia, USA
- 2014/09/12 **Nuclear matrix elements for neutrinoless double beta decay: multi-reference covariant DFT**, *The Autumn meeting of Chinese Physical Society (CPS2014)*, Sep. 11-14, 2014, Harbin, China
- 2014/09/ **Multi-reference covariant density functional theory for nuclear spectroscopy: recent progress**, *The long-term workshop "Present Status of the Nuclear Interaction Theory"*, Aug 25-Sep 19, 2014, Kavli Institute for Theoretical Physics China (KITPC), Chinese Academy of Sciences, Beijing
- 2011/10 **Impurity effect of Lambda hyperon on the collective excitation of atomic nuclei**, *The 18th Nuclear Physics Workshop "Nuclear Collective Phenomena "*, September 28 - October 02, 2011, Kazimierz Dolny, Poland

- 2011/09 **3D angular momentum restored calculations with a relativistic point-coupling Lagrangian**, *International workshop on "Restoring broken symmetries within the nuclear Energy Density Functional method"*, September 13-15, 2011, CEA/SPhN, Gif-sur-Yvette Cedex, Paris, France
- 2009/09 **An extended covariant density functional theory for low-lying states of exotic nuclei**, *The 2009 Autumn meeting of Chinese Physical Society*, September 17-20, 2009, Shanghai, China

Contributed talks in conferences/workshops/schools/seminars

- 2023/12/29 **Advances in Modeling medium-mass Λ hypernuclei with mean-field-based approaches**, *Seminar, Xiamen University, Dec 29, 2023*
- 2023/12/18 **Advances in Modeling neutrinoless double-beta decay with operators from chiral effective field theory**, *Seminar, the Few-body Systems in Physics Laboratory, RIKEN, Dec 18, 2023*
- 2023/12/8 **neutrinoless double-beta decay and nuclear many-body problems with operators from chiral effective field theory**, *The 1st winter school on theoretical physics, Guangzhou, Dec 4-9, 2023*
- 2023/10/26 **Multi-reference covariant density functional theory for low-lying states of odd-mass nuclei**, *Seminar, East China Normal University, Shanghai, Oct. 26, 2023*
- 2023/10/25 **Modeling neutrinoless double-beta decay with operators from chiral effective field theory**, *Seminar, Tongji University, Shanghai, Oct. 25, 2023*
- 2023/10/24 **Theory of neutrinoless double-beta decay in atomic nuclei**, *Lectures, Tsung-Dao Lee Institute at Shanghai Jiao Tong University, Oct. 24, 2023*
- 2023/10/23 **Modeling neutrinoless double-beta decay with operators from chiral effective field theory**, *Seminar, Shanghai Jiao Tong University, Shanghai, China*
- 2023/6/28 **Nuclear matrix elements for neutrinoless double-beta decay**, *The 1st Jiangmen Summer School on Neutrinos, June 20-29, 2023, Kaiping, China, link*
- 2023/5/15 **Recent progress in ab initio modeling nuclear matrix elements for neutrinoless double beta decay**, *18th National Conference on Nuclear Physics, Huzhou, China*
- 2023/4/22 **Computing atomic nuclei starting from nuclear chiral forces: Introduction**, *The 12th Workshop on CDFT in Nuclear Physics, Tianjin, April 22, 2023*
- 2022/06/21 **ab initio studies of medium-mass nuclei and the matrix elements of neutrinoless double beta decay**, *Seminar, Institute of Modern Physics, Lanzhou, China*
- 2018/11/15 **Computing low-lying states of deformed nuclei with chiral NN+3N interactions**, *FRIB/NSCL, Michigan State University, East Lansing, USA*
- 2018/11/19 **Beyond mean-field approaches for nuclear physics**, *Physics Colloquium, Western Michigan University, Kalamazoo, USA*
- 2018/11/15 **Computing low-lying states of deformed nuclei with chiral NN+3N interactions**, *FRIB/NSCL, Michigan State University, East Lansing, USA*
- 2021/05/21 **Ab initio calculation of nuclear matrix elements for neutrinoless double beta decay**, *May 21, 2021 Workshop of neutrinoless double beta decay, Zhuhai, China*

- 2018/09/07 **GCM-based IMSRG and neutrinoless double beta decay**, 7-8 September, 2018 DBD Collaboration Meeting at Lawrence Berkeley National Laboratory, USA
- 2018/05/29 **Multi-reference in-medium similarity renormalization group for deformed nuclei**, 2018 NUCLEI Collaboration Meeting, May 29 to June 1, 2018, Knoxville, Tennessee, USA
- 2018/04/03 **Generator Coordinate Method for Nuclear Low-Lying States: from MR-EDF to MR-IMSRG Calculations**, Theory Seminar, NSCL/FRIB at MSU, East Lansing, USA
- 2017/06/20 **Multi-Reference In-Medium SRG for Neutrinoless Double Beta Decay**, INT Program 17-2a, Neutrinoless Double-beta Decay, University of Washington, Seattle, USA
- 2017/06/08 **Multi-reference in medium SRG for neutrinoless double beta decay**, NUCLEI Collaboration meeting, June 6-8, 2017 in Santa Fe, NM, USA
- 2017/02/03 **Multi-Reference In-medium Similarity Renormalization Group for the Nuclear Matrix Elements of Neutrinoless Double Beta Decay**, 2017 DBD Collaboration Meeting, UMass Amherst, USA
- 2016/10/12 **Multi-reference covariant density functional theory for the nuclear matrix elements of Neutrinoless Double Beta Decay**, Sichuan University, Chengdu, China
- 2016/08/01 **Multi-reference In-medium SRG for the Nuclear Matrix Elements of Neutrinoless Double Beta Decay**, 2016 DBD Collaboration Meeting, August 1-2, 2016, FRIB, Michigan State University, East Lansing, USA.
- 2015/06/ **Towards Ab-initio Calculation of Nuclear Matrix Elements for Neutrinoless Double Beta Decay**, NUCLEI SciDAC Collaboration Meeting, June 6-10, at Argonne National Laboratory, USA
- 2015/05/18 **Building New Nuclear Theory Research at York**, University of York, UK
- 2015/02/12 **Beyond relativistic mean-field study of low-lying states for quadrupole-octupole deformed nuclei**, Collaboration workshop, Feb.12-13, 2015, Aizu University, Japan
- 2014/09/18 **Unveiling nuclear structure with spectroscopic methods**, School of Physics and Nuclear Energy Engineering, Beihang University, Beijing, China
- 2014/03/17 **A relativistic energy density functional calculation of the nuclear matrix elements in neutrinoless double beta decay**, International Molecule-type Workshop on New correlations in exotic nuclei and advances of theoretical models, Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Japan, 2014
- 2013/12/04 **Covariant density functional theory for nuclear collective excitations**, Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Japan
- 2013/12/03 **Beyond mean-field description of impurity effect of Lambda hyperon on nuclear collective excitations**, The Strangeness Nuclear Physics Laboratory, RIKEN Nishina center for Accelerator-Based Science, Japan

- 2013/10/09 **Description of nuclear collective excitations with multi-reference covariant density functional theory: Role of dynamical correlation effects**, *The Theoretical Nuclear Physics Laboratory, RIKEN Nishina center for Accelerator-Based Science, Japan*
- 2013/09/10 **Multi-Reference Covariant Density Functional Theory for Nuclear Spectroscopy**, *Sendai Nuclear Science Colloquium, Tohoku University, Japan*
- 2012/01/17 **Beyond mean-field study of low-lying collective excitation states in lead region**, *Workshop on Coulex analysis in lead region, Jan. 16-17, 2012, K.U. Leuven, Belgium*
- 2011/06 **Effects of triaxiality in low-lying states of magnesium isotopes: a relativistic 3DAMP+GCM study**, *International Symposium: "Advances in Nuclear Many-Body Theory", Primosten, Croatia, June 7-10, 2011*
- 2010/09 **Configuration mixing of angular momentum projected triaxial relativistic mean-field states**, *The 17th Nuclear Physics Workshop, "Marie & Pierre Curie", "Symmetry and symmetry breaking in nuclear physics", 22-26th September 2010 in Kazimierz Dolny, Poland*
- 2010/06 **Beyond the relativistic mean-field theory: configuration mixing of three-dimensional angular momentum projected states**, *The BLTP/JINR - KLFTP/CAS Joint Workshop on NUCLEAR PHYSICS, Dubna, Russia, June 28-July 4, 2010*
- 2009/06 **Extending the covariant density functional theory for nuclear low-lying excited states – Three-dimensional angular momentum projected generator coordinate method**, *"Relativistic many-body problems for heavy and superheavy nuclei", Beijing, KITPC/ITP-CAS, June, 2009*